Outstanding

GT Sheet 2

1\. a) Labour maximises utility choosing x, for which FOC is

$$\frac{\partial u_L}{\partial x} = \frac{(x+y)-x}{(x+y)^2} - 1 \overset{!}{=} 0$$

$$\Rightarrow (x+y)^2 = y.$$

$\pm\sqrt{y} - y = x$ but $x, y > 0$ so

$$B_L(y) = \sqrt{y} - y.$$

Symmetrically, $B_C(x) = \sqrt{x} - x$.

b) A NE is a strategy profile where each player plays a BR against the opponent's strategy. [In exams for GT, do you still need to do this defining part?]

Depends on the question. Some will specifically ask you to define an eq$^{\underline{m}}$ concept. Other, you can just start solving.

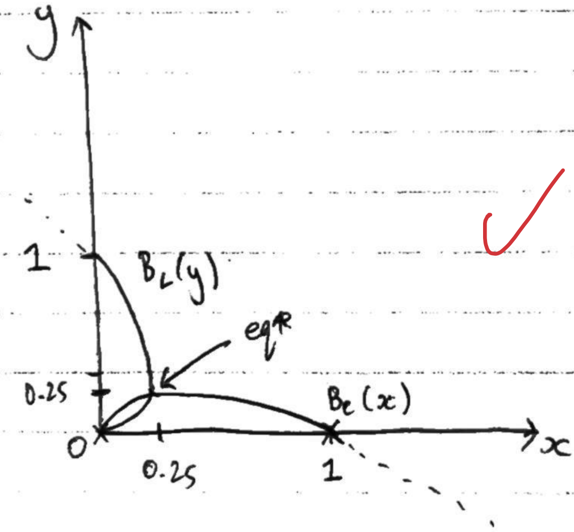
Suppose Labour spends x^*, then Conservatives spend $\sqrt{x^*} - x^*$; we require $x^* = \sqrt{x^*} - x^*$.

$\Rightarrow 4x^{*2} = x^*$; $x^* \neq 0$ so $x^* = 0.25$

and symmetrically $y^* = 0.25$

So $(0.25, 0.25)$ is NE with equal vote shares and payoffs $(0.25, 0.25)$

c)



d) The election would be tied, and neither would've spent any money on advertising. So we should define $u_i = \frac{1}{2}$ here for $i \in \{L, C\}$. But this isn't a NE; each player could profitably deviate by spending ε on advertising to win 100% of the vote share and have higher utility, $1-\varepsilon$.

e) Assuming x, y must be nonnegative, for (0,0) to be NE it needs to be the case -amongst other things- that $u_i(0+\varepsilon, 0) \le u_i(0,0)$, for small ε.

✓ But $u_i(0,+\varepsilon, 0) \to 1$ for small ε. So we must set $u_i(0,0) \ge 1$.
~~But that~~ This guarantees (0,0) is NE since $\forall x>0\ \forall y>0,\ \frac{x}{x+y} < 1$.
So
But that implies spending zero on advertising and tying the election is the most preferred outcome, which doesn't make sense as opposed to, say, spending ε and winning.

2 a) ~~$BR_A(b)$~~ The payoffs for A are:

✓
$$\begin{cases} X - a & \text{if } a > b \\ \frac{1}{2}X - a & \text{if } a = b \\ -a & \text{otherwise} \end{cases}$$
, where a, b are evidence from A, B.

~~If~~ The best response $BR_A(b)$ is

$$\begin{array}{ll} \text{undercutting} \rightarrow & \begin{cases} b+\varepsilon & \text{if } b < X \\ 0 & \text{otherwise} \end{cases} \\ \text{avoid pointless losses.} \rightarrow & \end{array}$$

i.e. $\emptyset$

← (not strictly defined as one best response, but infimum is b)

indeed

b) PSNE is found at intersection of BR curves. But given that $X > 0$, the BRs do not cross.

✓
- (0,0) is not a PSNE since each player can do better by paying ε evidence.
- There is no BR to any evidence amounts $< X$.
- And evidence $> X$ is a never-best response, so amounts cannot be in NE.
- Also (X, X) isn't NE since BR to X is 0.

So there's no NE.

[More concise way to show this?]

Could draw a picture.

Specifically, P(win | opp played c_0) goes from 1/2 to 1. So P(win) increases by $\frac{1}{2}p$ where ~~max~~ $p = P(\text{contribution} = c_0)$

c). ~~At~~ Suppose there's an atom at c_0. Then with positive probability both oligarchs contribute c_0 and tie. But if a player deviated by paying $c_0 + \varepsilon$, their expected payoff would increase, since the Δ in evidence cost is arbitrarily small but ~~ex~~ P(winning) increases by a discrete amount.

Excellent

So c_0 isn't ~~strictly dominated~~ a best response, by small upward deviations. But then it can't be part of an equilibrium mixed strategy, since symmetry means the two oligarchs ~~must~~ ~~for~~ have the same CDF.

[Wait, but even though game is symmetric, eq* could be asymmetric. So how do we rule out atoms at c_0? e.g. if my opponent has an atom, it's not BR for me to have one – but that doesn't preclude them from having one!]

I think the question is implicitly asking for a symmetric NE.

d). Suppose there's a gap in B's support ~~between~~ (g_l, g_h). ~~Then this~~ ~~BRs for any~~ ~~Then for A, it's never a BR to make contributions in (~~

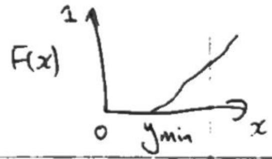
This means the CDF is flat on that interval.

By indifference principle, A must get same expected payoff from all actions they mix over. ~~Both~~ ~~(after symmetry)~~ But g_l and g_h are contributions with exactly equal P(winning), and g_h costs more. So g_h is strictly dispreferred to g_l, which is a contradiction.

$= 1 - F(g_l) = 1 - F(g_h)$.

[I guess unless g_l and g_h also are never played, so don't need to be indifferent about them?]

you took g_l & g_h to be in the support.



e) (Contributions must be nonnegative.) Suppose $y_{min} > 0$. ~~Then since it's in the support of playing with reward the strictly dominated an eq* strategy - there can't be profitable~~ If you play y_{min}, you're guaranteed to lose (probability of tie is zero) and get payoff $-y_{min}$. But this isn't ever a best response, since playing 0 with payoff 0 is better. So y_{min} couldn't be in the support. #

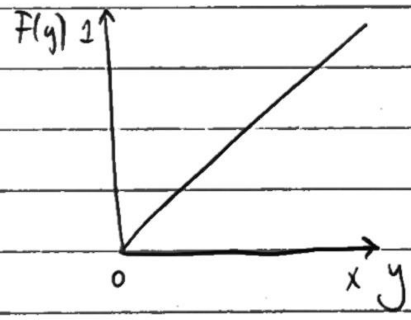
f) Expected payoff of every contribution in support must be equal.

Since ~~zero~~ $y=0$ never wins, $v(0) = 0$.

$$\text{So } v(y) = 0 = (x-y)\cdot \underbrace{F(y)}_{P(\text{win})} - y\cdot \underbrace{(1-F(y))}_{P(\text{lose})}$$

$$F(y) = \frac{y}{x}, \quad \text{for } y \in [0, x]$$

Nice



i.e. $y \sim U[0, x]$

	L	R
T	6,6	2,8
B	8,2	0,0

3 a) • PSNE by underlining BR payoffs:

◦ (B, L) and (T, R)
(8, 2) (2, 8)

• MSNE:

◦ None where only one player A mixes, since A is not indifferent between actions given B plays a pure strat.

◦ Let T, L be played w.p. p, q. For Row by indifference principle:

$$6q + 2(1-q) = 8q$$

and for Col, $6p + 2(1-p) = 8p$.

So $p = q = 1/2$. MSNE is $(\frac{1}{2}T + \frac{1}{2}B, \frac{1}{2}L + \frac{1}{2}R)$

with expected payoffs $(4, 4)$

b) Suppose I'm player 1. Then if my opponent adopts this strategy, we want to see if me also adopting is a BR.

Player 2	R	L
P2 BR	T	B
	↑ heads	↑ tails

In either case of how the coinflip goes, the BR to the action my ~~coinflip~~ opponent is directed to take is what I'm directed to take.

So yes, I'm willing to follow the instructions. (Symmetric for 2)

We get $\mathbb{E}$ payoffs $(5, 5)$ which are higher than the MSNE. Welfare is the same as either PSNE; this correlating device (sunspot) allows us to randomly select between the two ~~pure~~ PSNE; as expected $\mathbb{E}$ payoffs are a convex comb. (midpoint here, since outcomes equiprobable).

c) Here, we randomise between (T, L), (T, R) and (B, L), using correlated but private signals. Partitions are $(\{x, y\}, \{z\})$ and $(\{x\}, \{y, z\})$ for P1 and P2 respectively.

$\sigma_1(\{x, y\}) = T$, $\sigma_1(\{z\}) = B$; $\sigma_2(\{x\}) = R$; $\sigma_2(\{y, z\}) = L$.

- As P1 if I observe $\{x, y\}$ then $v_1(T) = \frac{1}{2}6 + \frac{1}{2}2 \geq \frac{1}{2}4 = v_1(U)$ and if I observe $\{z\}$ then $v_1(B) = 8 \geq 8 = v_1(T)$

So I'm weakly willing to follow / no strict incentive to deviate.

And symmetric for P2.

$\mathbb{E}$ payoffs are $(16/3, 16/3)$ which are better than in b); welfare higher than any NE.

This CE allows for coordination to avoid the bad outcome (B, R).

Note that each player still has a PSNE they prefer, though.

d) Yes, it seems quite reasonable. ~~PS~~NE are equivalent to CE where each player receives and independent signal; in real life there are often signals available for coordination which are in fact correlated, so generalising NE to allow for this is sensible.

Public signals / sunspots are especially intuitive – there are many common-knowledge norms we have (like the meaning of traffic lights) used to coordinate.

4. $p = 1 - q_1 - q_2$.

1 \ 2	C_L	C_H
C_L	α	$\frac{1}{2} - \alpha$
C_H	$\frac{1}{2} - \alpha$	α

$\alpha \in [0, \frac{1}{2}]$

a) Start with $i = 1$, $j = L$. $P(c_2 = L \mid c_1 = L) = 2\alpha$. $\left(= \frac{\alpha}{\alpha + \frac{1}{2} - \alpha}\right)$

More generally, firm i's posterior is that ~~firm~~ $-i$ has the same cost w.p. 2α and different cost w.p. $1 - 2\alpha$.

b) A strategy is a complete contingent plan of action. Let (q^i_L, q^i_H) denote the strategy for a firm, i, where $q^i_L, q^i_H \in \mathbb{R}^{\geq 0}$ represent production output in the case of being low or high cost respectively.

(Or, if we need mixing to be possible, a strategy is a tuple of probability distributions ~~with support~~ defined on $x \geq 0$)

Yes! Good.

The expected payoff for firm i with realised cost c_j is

$$\mathbb{E}\left[(1 - q_i - q_{-i} - c_j) \cdot q_i \mid c^i = c_j\right]$$

But since q_{-i} is not known ex ante, we need to use i's posterior to simplify the $\mathbb{E}[q_{-i}]$ term.

$$\mathbb{E}\left[q_{-i} \mid q_m^{c_i=j}\right] = 2\alpha q_j^{-i} + (1-2\alpha)\, q_{-j}^{-i}$$ where $-j$ denotes "different cost".

So ~~$\pi_i(q_i) =$~~

$$\pi_i(q_i, q_H^{-i}, q_L^{-i}) = \left(1 - q_i - 2\alpha q_j^{-i} - (1-2\alpha)\, q_{-j}^{-i} - c_i\right) \cdot q_i$$

c) We want to maximise interim expected profits, for which the FOC is $\frac{\partial \pi_i}{\partial q_i} \overset{!}{=} 0$.

Here, we have $1 - 2q_i - 2\alpha q_j^{-i} - (1-2\alpha) q_{-j}^{-i} - c_i = 0$

$$\Rightarrow q_i^j = \frac{1 - 2\alpha q_j^{-i} - (1-2\alpha) q_{-j}^{-i} - c_i}{2}$$

Or, more explicitly:

$$q_L^i = \frac{1 - 2\alpha q_L^{-i} - (1-2\alpha) q_H^{-i} - c_L}{2}$$

and $$q_H^i = \frac{1 - 2\alpha q_H^{-i} - (1-2\alpha) q_L^{-i} - c_H}{2}$$

In BNE, each player-type plays a best responce to their posterior belief of opponent's strategy. [? - unsure about how to generalise normal NE statement]

Not sure what you mean here.

Given there's only one ("the") BNE and this is a symmetric game, we can impose symmetry. [Assume this holds in BNE too?]

ok

So $q_L^i = q_L^{-i} = q_L$; same for H.

$$\therefore q_L = \frac{1 - 2\alpha q_L - (1-2\alpha) q_H - c_L}{2} \quad (1)$$

$$q_H = \frac{1 - 2\alpha q_H - (1-2\alpha) q_L - c_H}{2} \quad (2)$$

Rearrange (1): $q_L = \dfrac{1-(1-2\alpha)q_H - c_L}{2\alpha+2}$ (1')

(1') into (2): $(2\alpha+2)q_H = 1 - c_H - (1-2\alpha)\dfrac{1-(1-2\alpha)q_H - c_L}{2\alpha+2}$

$$\underbrace{\left[(2\alpha+2)^2 - (1-2\alpha)^2\right]}_{(\cancel{2\alpha}+2+1-\cancel{2\alpha})(2\alpha+2-1+2\alpha) \;=\; 3(4\alpha+1)} q_H = \underbrace{(2\alpha+2)(1-c_H)}_{2\alpha+2-2c_H-2\alpha c_H} - \underbrace{(1-2\alpha)(1-c_L)}_{-1+c_L+2\alpha-\cancel{2\alpha c_L}}$$

$q_H = 4\alpha - 2\alpha(c_L+c_H)$

$$q_H = \frac{4\alpha - 2\alpha(c_L+c_H) - 2c_H + c_L + 1}{3(4\alpha+1)}$$

and similarly (just let $j' = -j$)

$$q_L = \frac{4\alpha - 2\alpha(c_L+c_H) - 2c_L + c_H + 1}{3(4\alpha+2)}$$

Nice

d) When $\alpha = 0$, $q_j = \dfrac{-2c_j + c_{-j} + 1}{3} = \dfrac{2 - 4c_j + 2c_{-j}}{6}$

$\alpha = \frac{1}{4}$, $q_j = \dfrac{2 - \frac{5}{2}c_j + \frac{1}{2}c_{-j}}{6}$

$\alpha = \frac{1}{2}$, $q_j = \dfrac{3 - 3c_j}{9} = \dfrac{2-2c_j}{6}$

For $\alpha = 1/2$, each firm is guaranteed identical costs. So c_{-j} is irrelevant; c^{-i} is known to be c^i.

At $\alpha = 0$, it's known that the costs are perfectly anticorrelated. No uncertainty.

$\alpha = 1/4$ means costs uncorrelated; $1/4$ prob of each state.

In general, quantity* is larger when your costs are lower and opponent's higher, as expected. $\alpha > 1/4$ gives increasingly strong evidence that your opponent's costs are high when yours are (+ vice versa; same mutatis mutandis for $\alpha < 1/4$). So, a high-cost firm expects its opponent to be high-cost and produce less, meaning they produce more (strategic subs.).